

core counts," informs Erraguntla. You can find more details at <http://techresearch.intel.com/articles/Tera-Scale/1421.htm>.

More than desktops and supercomputers

Multi-core processors are becoming an indispensable aspect of virtualisation too, especially in data centres.

Vamsi Krishna explains, "In the near future, multi-core processors will enable data centres to accomplish tasks more quickly and with greater energy efficiency. Areas such as virtualisation are also primed for a boost by multi-core processors. As AMD moves to eight- and 12-core platforms, far more virtual machines can be packed onto each physical server. By focusing heavily on power efficiency and virtualisation capabilities, businesses can now add performance and efficiency to their business without a significant cost."

R. Ravichandran, director-sales, Intel South Asia, says, "Industry reports point out that there will be approximately 10-15 billion devices in the next four to five years, and most devices like TVs, embedded devices and other consumer electronic devices (beyond traditional desktops, notebooks and servers) will have a link to the Internet. Given that there will be a proliferation of devices in the computing continuum, with rich media and video as killer applications, some of the handhelds and smart phones will need great computing capabilities that are energy-efficient... and multi-core will also pervade these segments."

Ravichandran cites the following examples to prove his point:

- **Safer and smarter roads:** A number of traffic accidents are caused by worn-down car tyres. Intel, along with industry players Kontron and ProContour, has developed a tyre-tread-monitoring embedded technology that is making roads smarter. Kontron, a member of the Intel Embedded and Communications Alliance, has developed a camera with an Intel Core2Duo processor that captures tyre-tread depth as a tyre passes over a specialised grate. This technology can alert drivers when their tyres need replacement to avoid potentially dangerous tyre blow-outs.
- **Medical imaging with multiple cores:** Physicians today collect more complex imagery of their patients than ever before. In order to accurately diagnose diseases and develop treatment strategies in a minimally-invasive manner, new imaging modes, methods and hardware are needed.

In collaboration with the Mayo Clinic, Intel has presented a paper titled Mapping High-Fidelity Volume Rendering for Medical Imaging to CPU, GPU and Many-Core Architectures, outlining how medical imaging benefits from

Multi-core microcontrollers: Powering your car, phone and more

Multi-core technology is no longer confined to high-performance computing or even desktop computing. It has entered the devices that we use every day, in the form of multi-core microcontrollers that are being extensively used in embedded systems.

Several companies, including Renesas Technologies, LSI Corporation, STMicroelectronics and Freescale Semiconductor, have introduced multi-core microcontrollers in the recent past. While the multi-core chips initially introduced for embedded systems were aimed at image processing and other multimedia products with heavy processing loads, recent products are targeted at a broad range of general-purpose devices. These facilitate the true real-time response times required by real-time control systems, safety-critical applications, etc, thanks to their ability to process instructions in parallel.

parallel processing in the Intel micro-architecture codenamed Nehalem. Medical volumetric imaging requires high-fidelity, high-performance rendering algorithms. They have now achieved performance improvements of more than one order of magnitude on a number of large 3-D medical datasets.

What to expect

Look into any device a few years down the line and you will be sure to find a multi-core processor in it. Specialists in the embedded arena, including ARM and RIM, are all launching multi-core models of their processors too.

Multi-core processors have already widened their ambit from supercomputers to desktops, mobiles and data centres. However, the real success and sustainability of the multi-core concept depends on whether it will be ably supported on the software front too, with a proper understanding and execution of true parallel programming principles. Considering the efforts being made by industry leaders to train developers on parallel techniques, this hurdle will be overcome soon, and multi-cores will be adopted even more rapidly. 

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